

The background is a complex, abstract composition. It features a low-poly, crystalline structure in shades of orange, red, and blue. Overlaid on this are several sets of concentric circles, some solid and some dotted, in a light gray or white color. The overall effect is a modern, technological aesthetic.

Natural Language Technology

Week 1 Introduction

Eliza

A First Conversational Technology

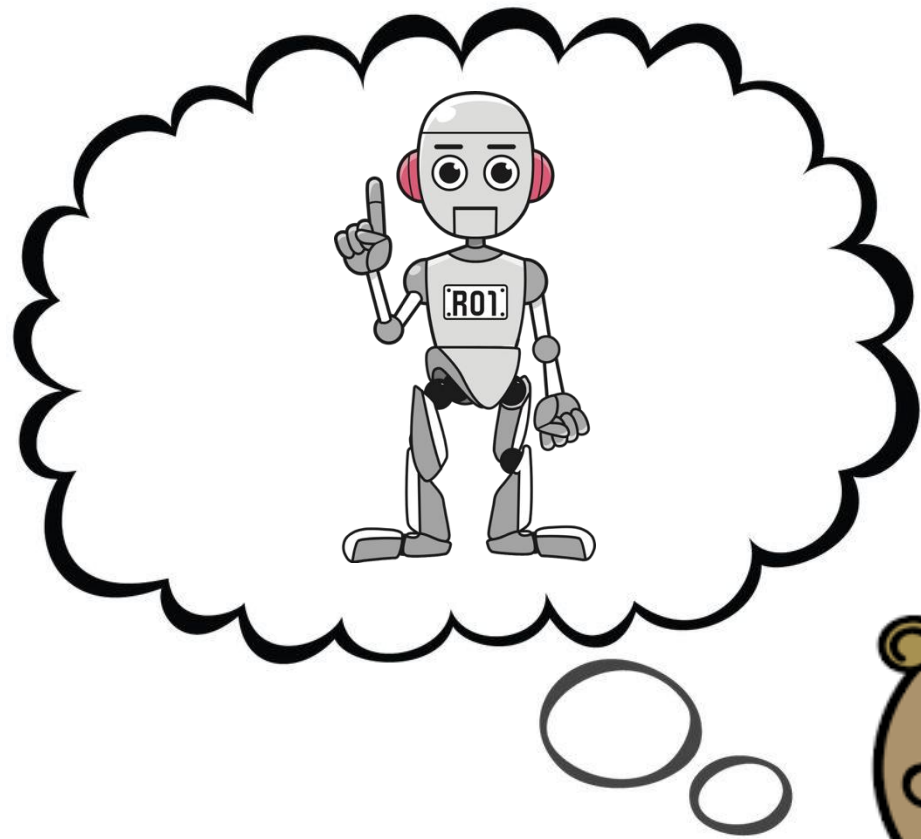
Joseph Weizenbaum, 1966



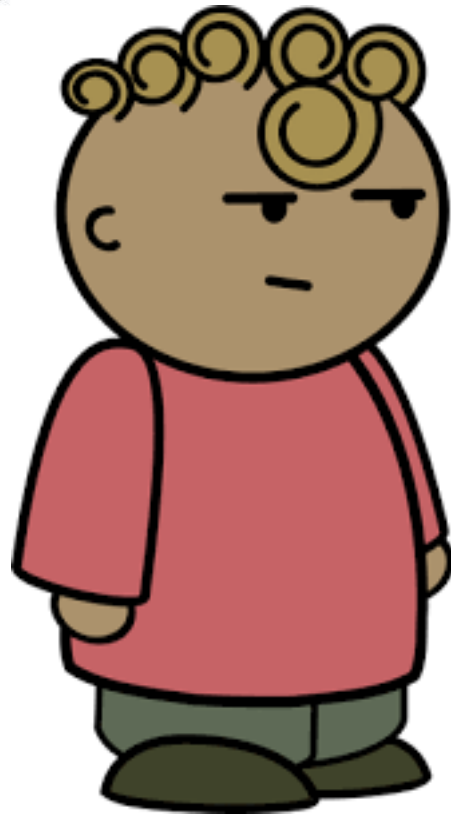
```
EE      LL      II      ZZ      AA      AA
EEEEEE  LL      II      ZZZ  AAAAAAA
EE      LL      II      ZZ   AA      AA
EEEEEE  LLLLLL  IIII  ZZZZZZZ  AA      AA
```

Eliza is a mock Rogerian psychotherapist.
The original program was described by Joseph Weizenbaum in 1966.
This implementation by Norbert Landsteiner 2005.

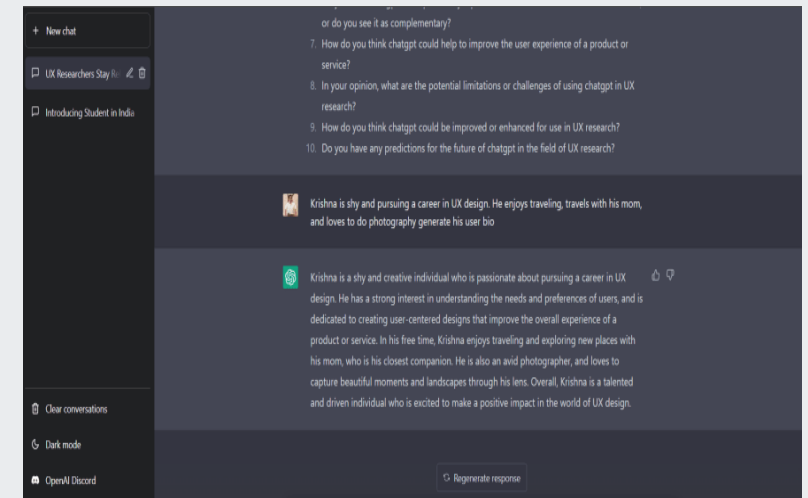
```
ELIZA: Is something troubling you ?
YOU:   Men are all alike.
ELIZA: What is the connection, do you suppose ?
YOU:   They're always bugging us about something or other.
ELIZA: Can you think of a specific example ?
```



The Eliza Effect



3 Contemporary Conversational Systems





Woebot

write ONE of the anxious thoughts that's bothering you here: ✎

Something bad is going to happen

write another one here. Try to keep them brief.. ✎

I could die in a plane crash

and one last one: ✎

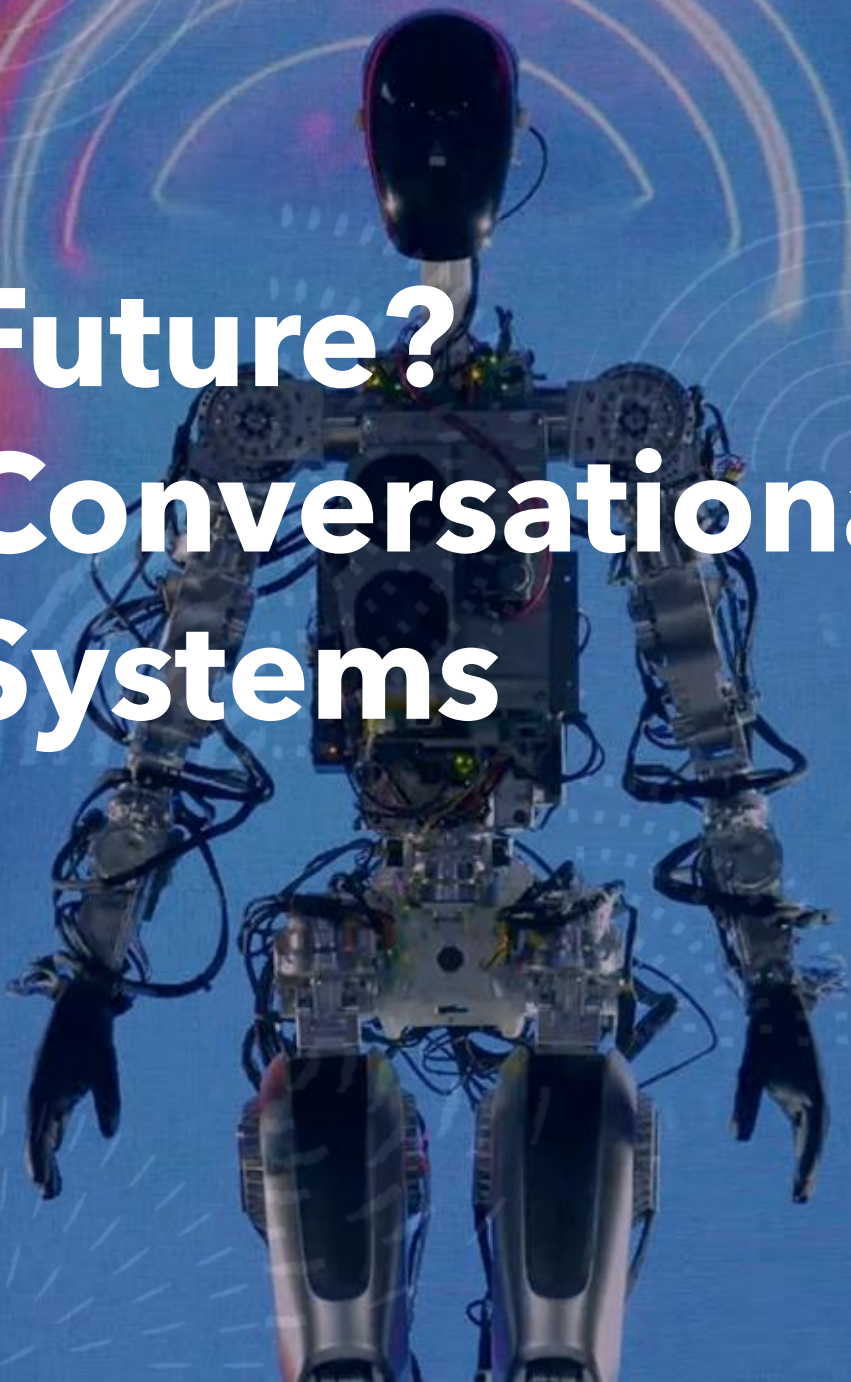
It will be terrifying knowing

Does your thought "I could die in a plane crash" have some catastrophizing in it?

catastrophizing?

this is when we put a negative spin on a situation

Future? Conversational Systems





The Basics

How do we
build
Conversational
Systems?

The ML Playground

Putting Deep
Learning and
AI to Work

Personalisation

Keeping the
User Engaged



The Basics

How do we
build
Conversational
Systems?

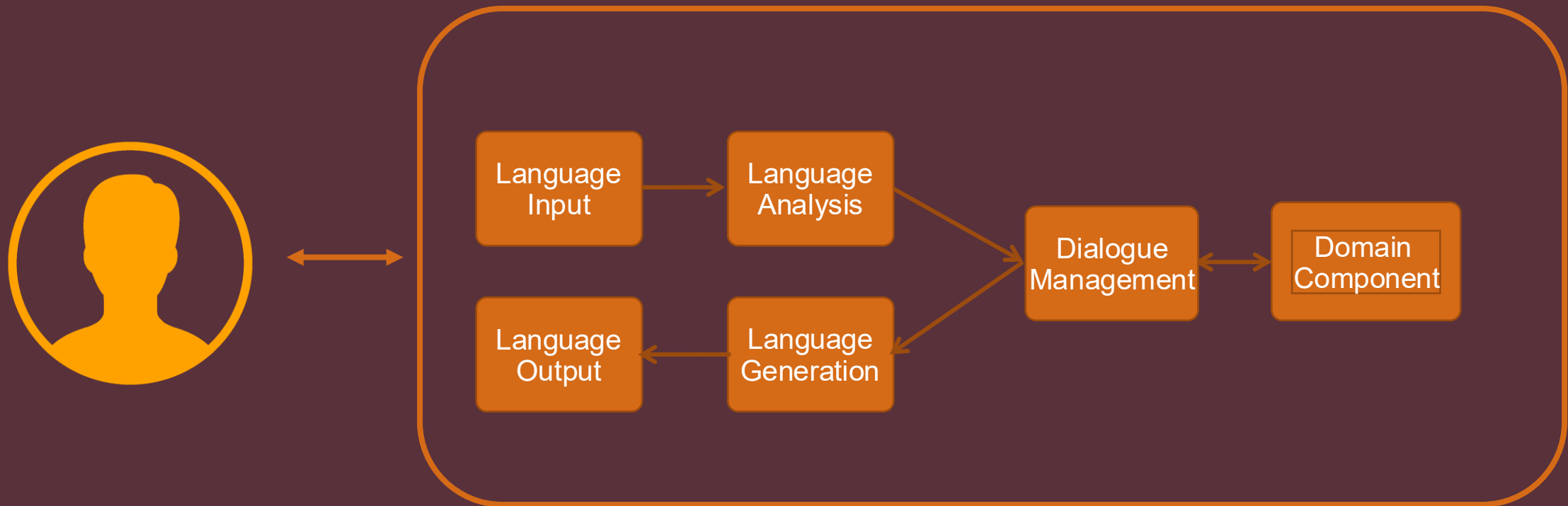
The ML Playground

Putting Deep
Learning and
AI to Work

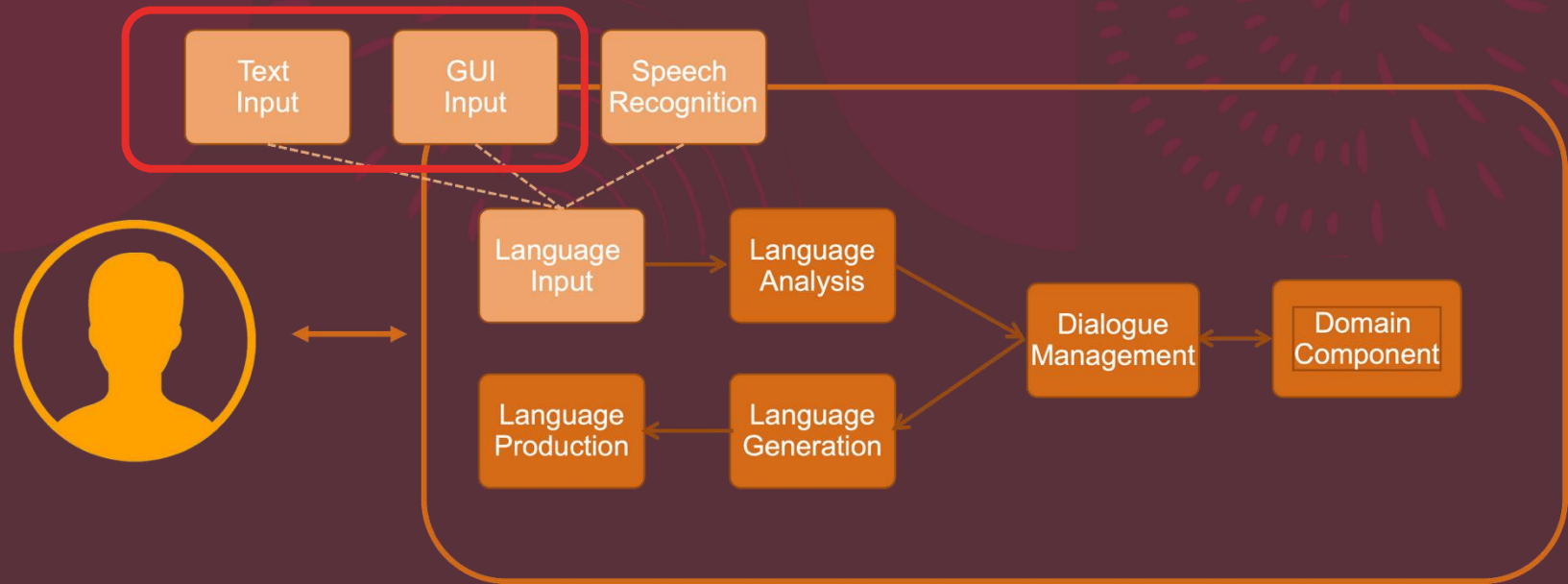
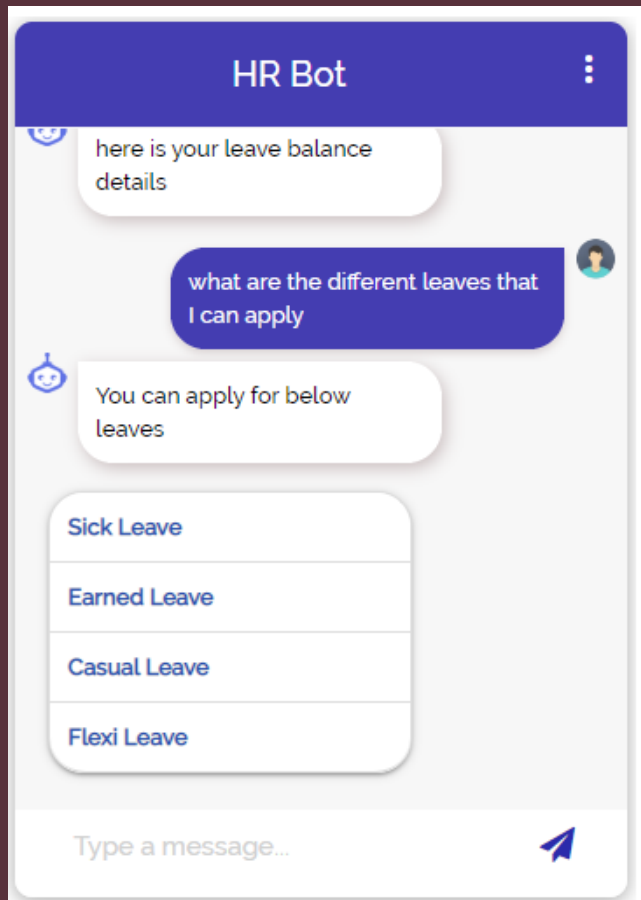
Personalisation

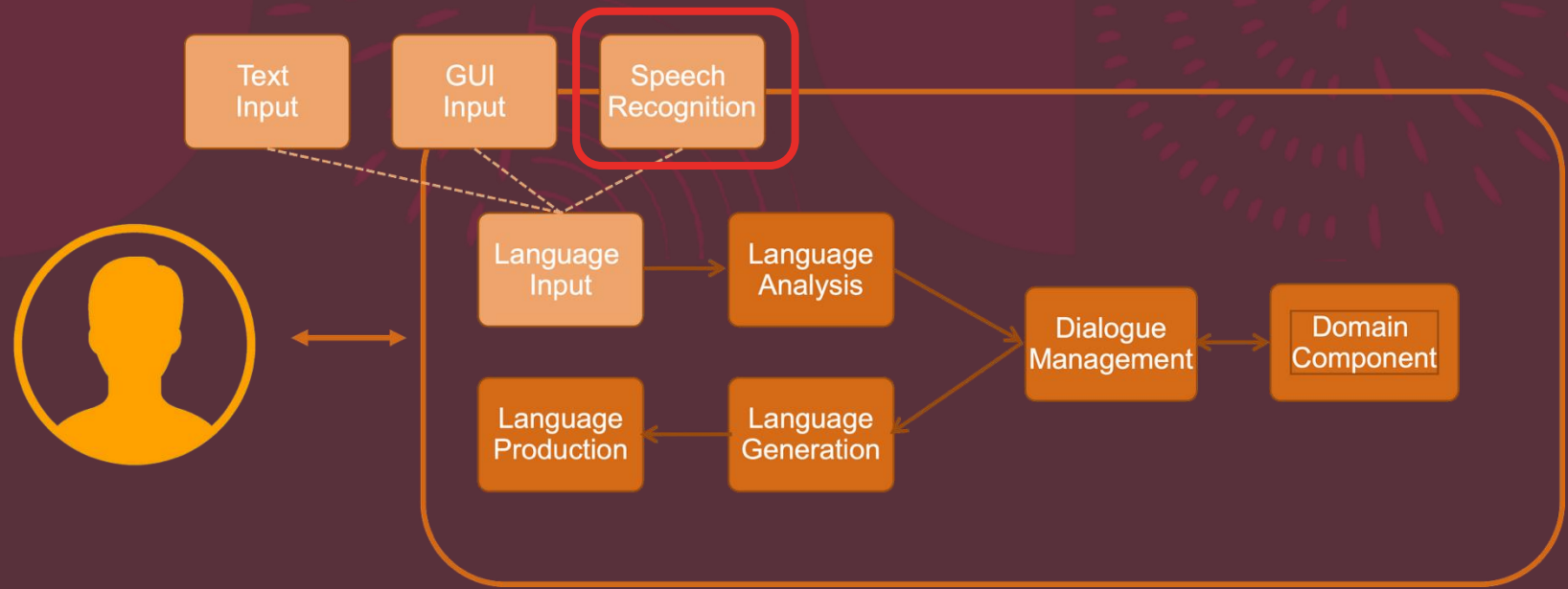
Keeping the
User Engaged

Conversational Architectures aka Dialogue Systems



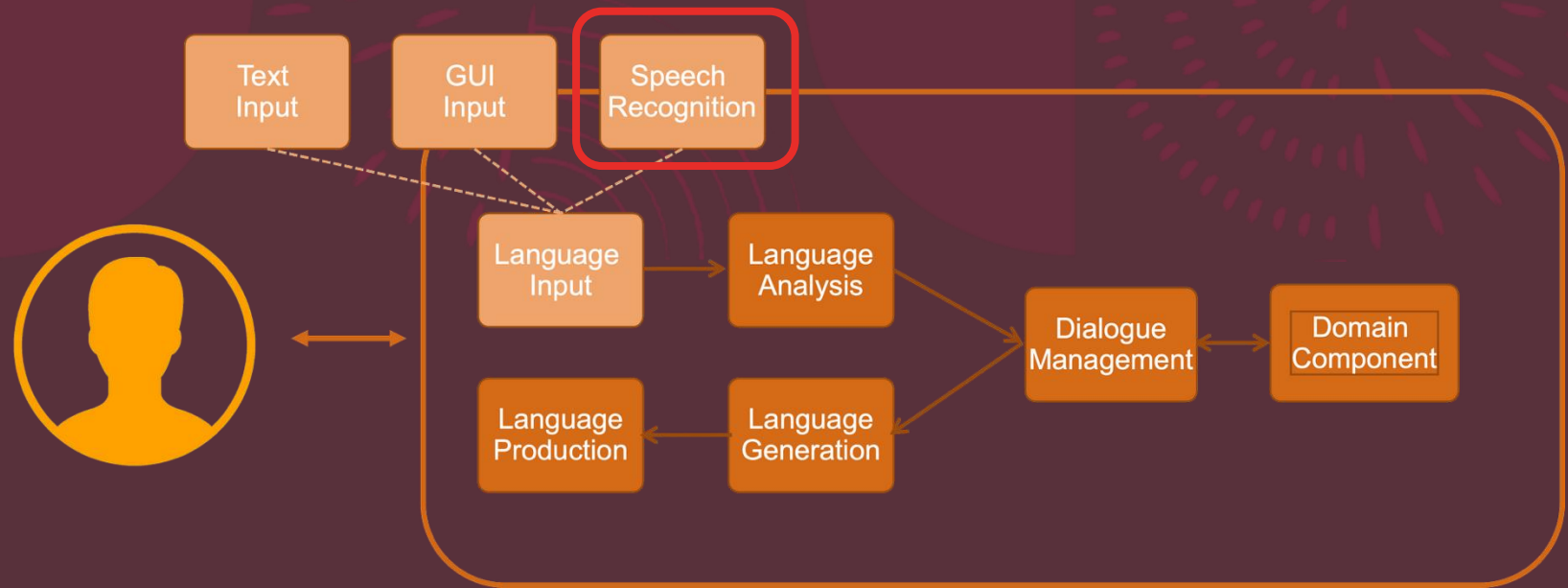
Architectures of natural language technology components used to enable conversational applications



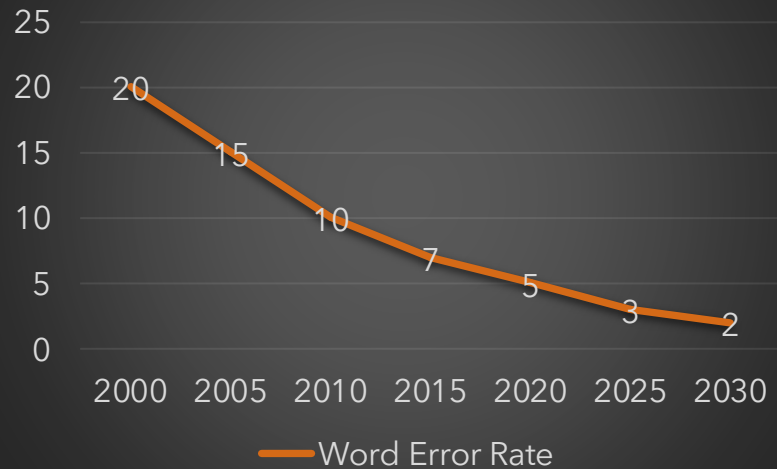


Technology Backbone:

- *Historically Hidden Markov Models*
- *Modern Systems based Recurrent Networks and Transformers*

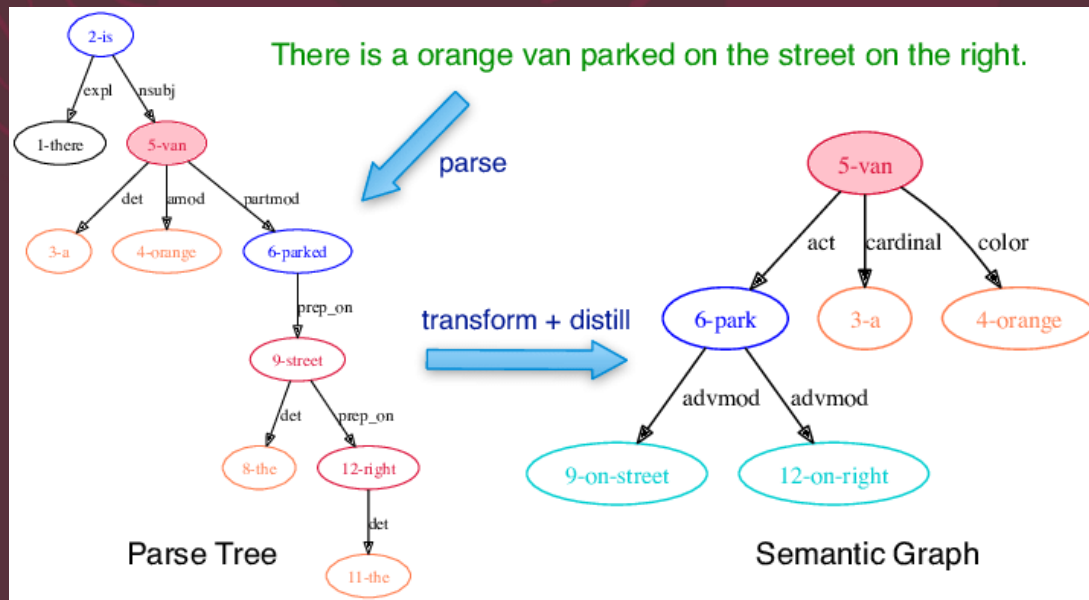


Word Error Rate

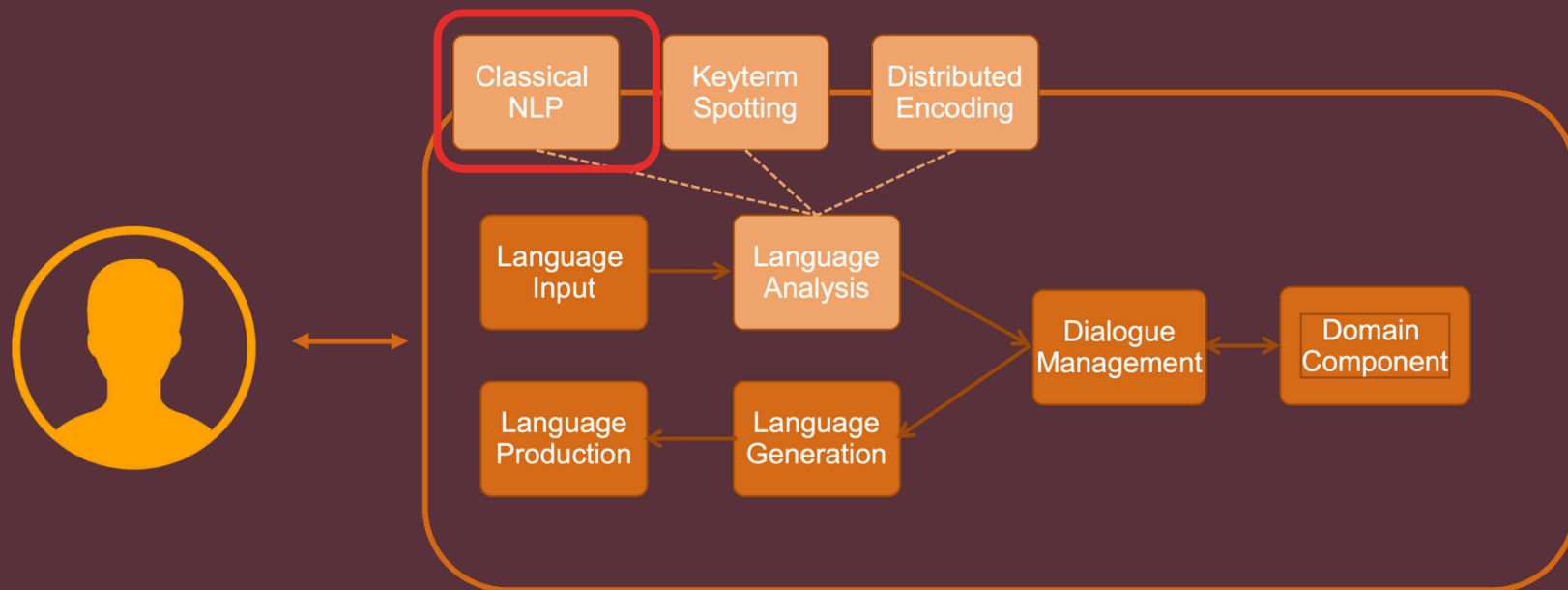


Challenges:

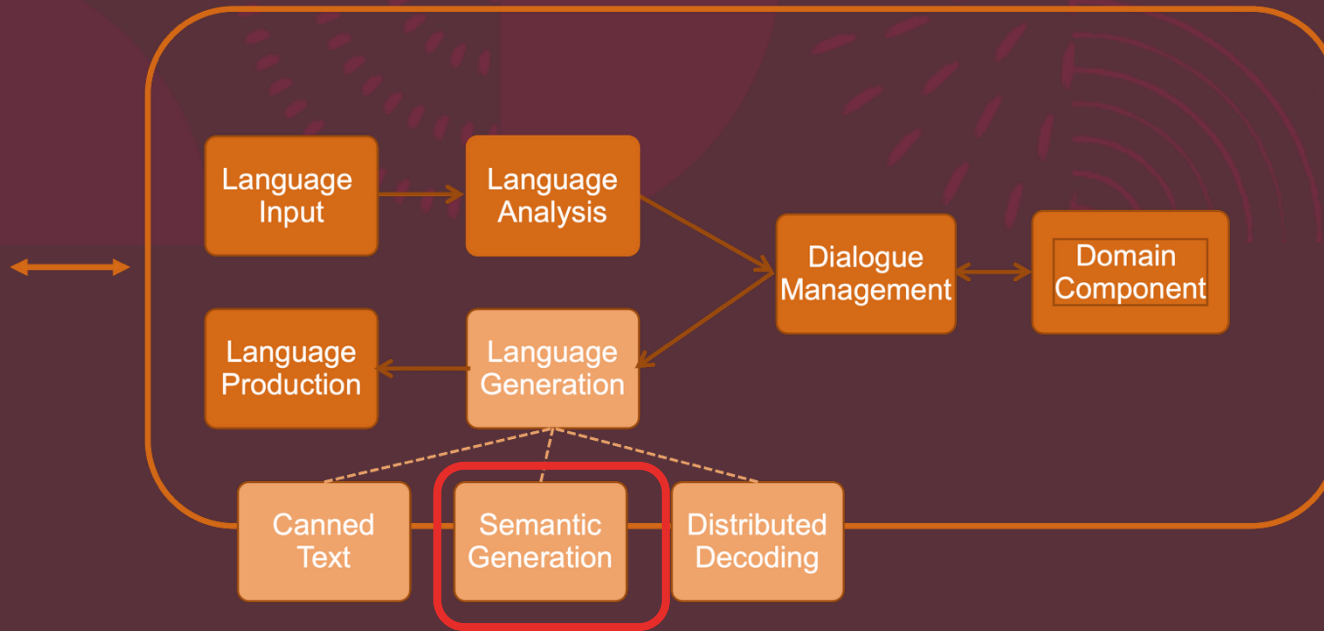
- *Beyond outdated error metrics*
- *Noisy Environments and multi-user settings*



https://www.researchgate.net/publication/264975659_Visual_Semantic_Search_Retrieving_Videos_via_Complex_Textual_Queries





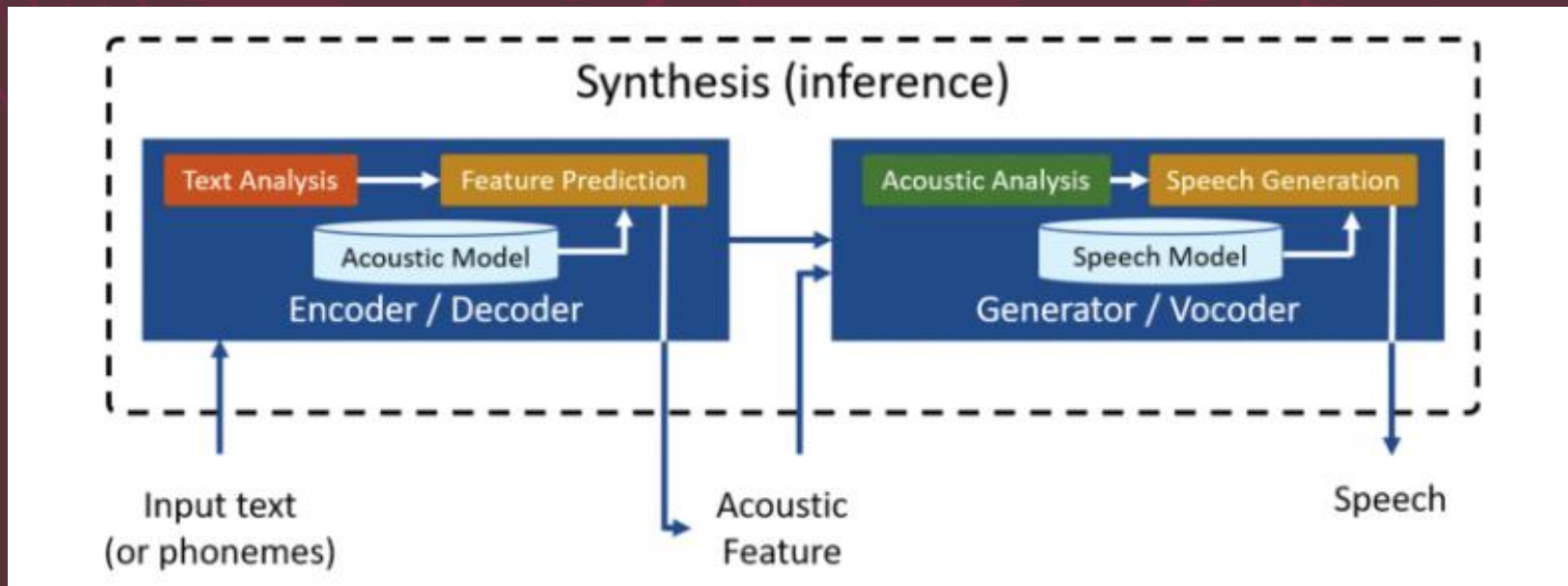


```
(10 / spatial-locating
:speechact (a0 / assertion :polarity positive
                        :speaking-time t0)

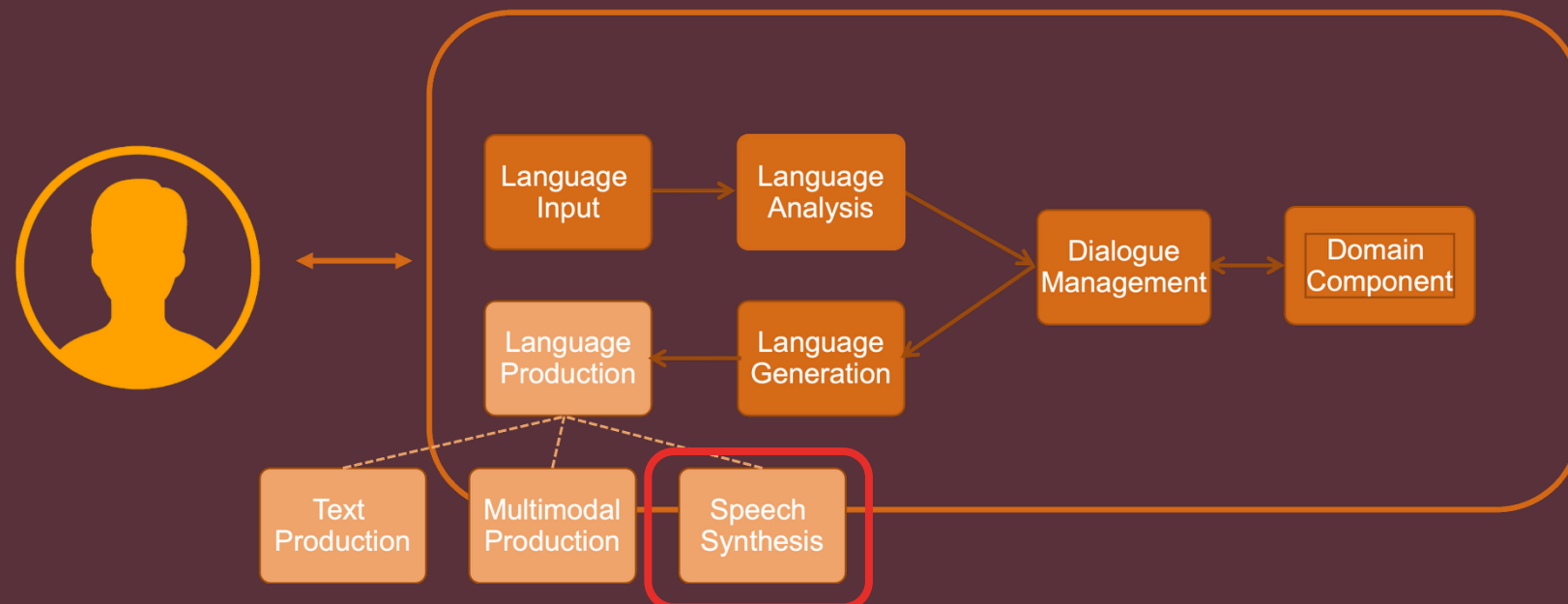
:reference-time-id t0
:event-time (t0 / time)
:theme d0
:domain (d0 / object :lex dog
          :identifiability-q notidentifiable)
:range (p0 / three-d-location :lex park
        :identifiability-q identifiable))
```

*Classical Semantic
Generation Systems*

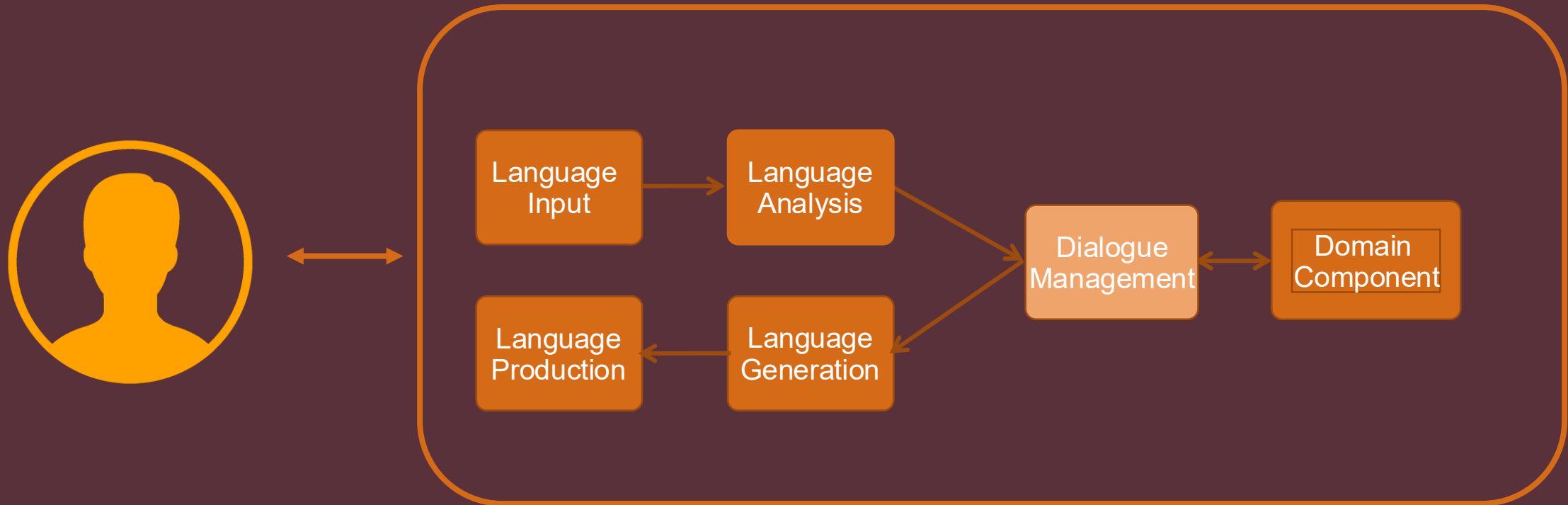
There is a dog in the park



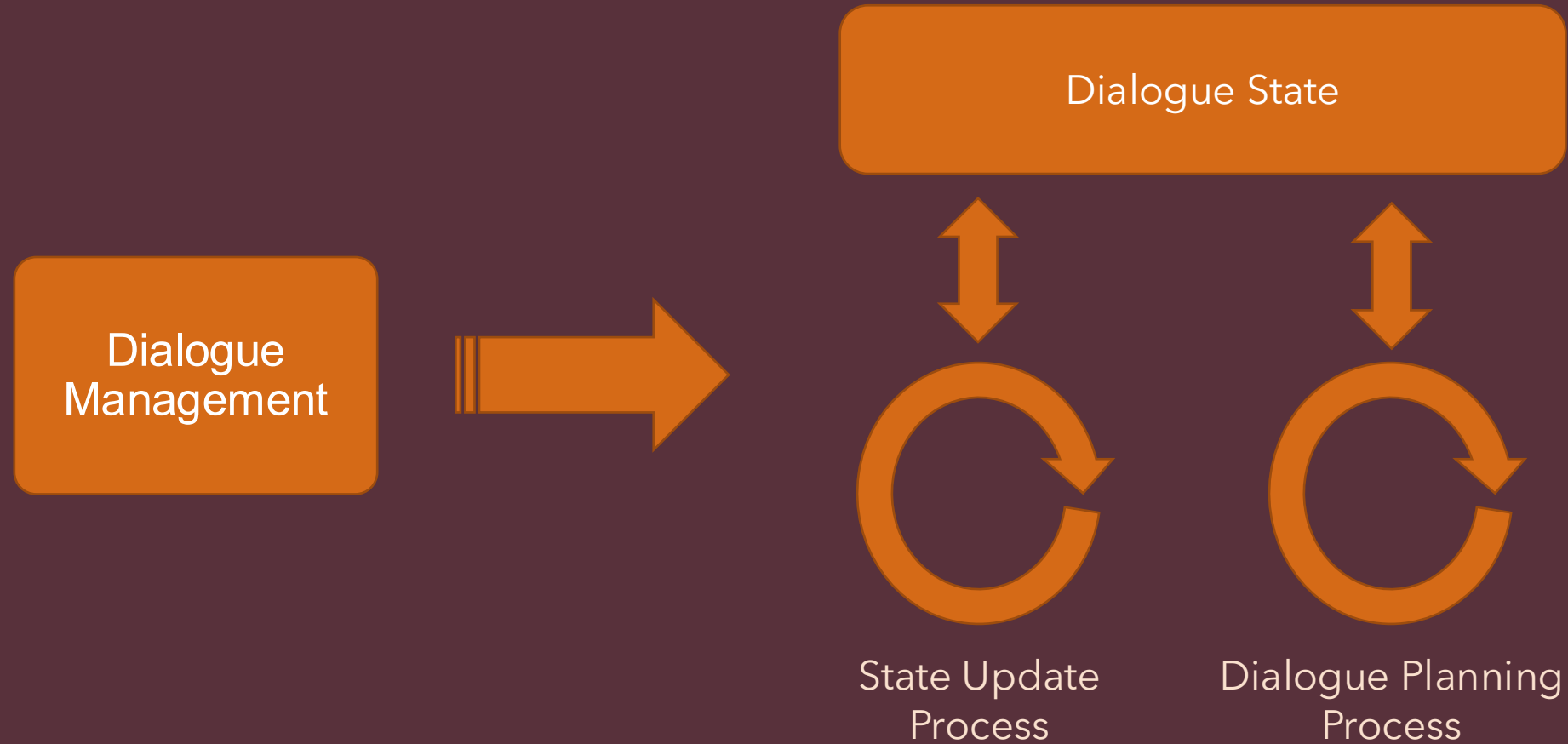
<https://towardsdatascience.com/state-of-the-art-of-speech-synthesis-at-the-end-of-may-2021-6ace4fd512f2>



Dialogue Management



State Management vs. Planning



Dialogue Manager Paradigms ++

Slot Based
Dialogue Systems

State Graph
Dialogue Systems

ChatBot Based
(ChatGPT)

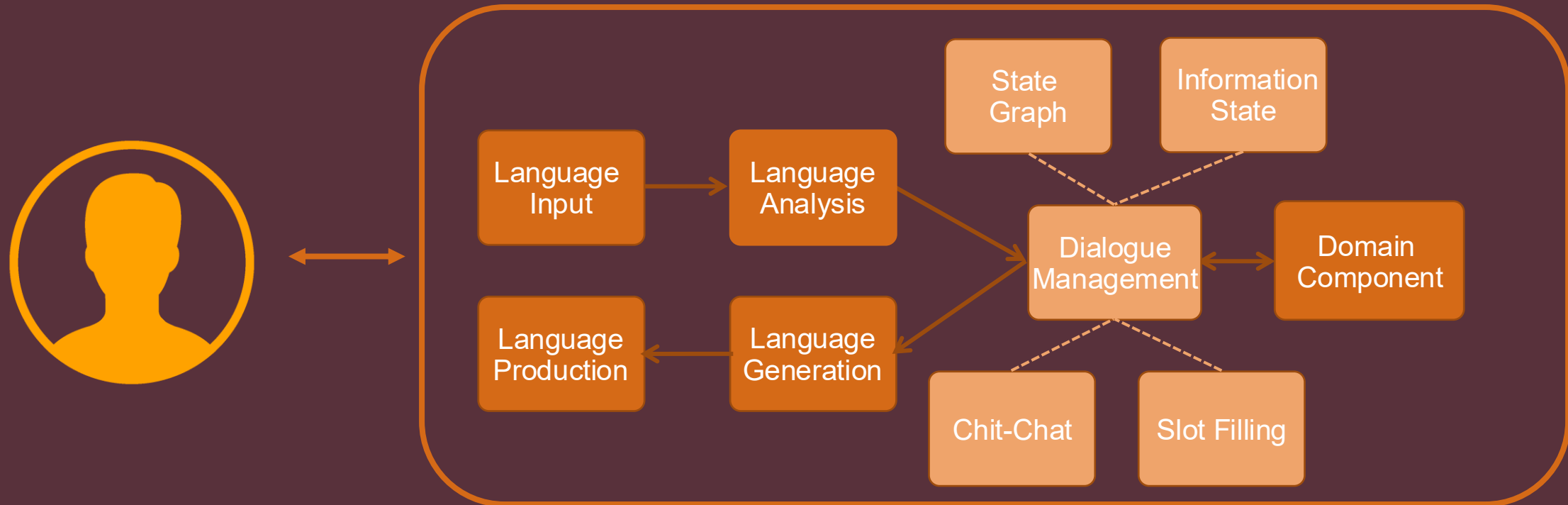
Agent Based
Dialogue Systems

Pull all 4 together
into Agentic LLM
solutions

Well Defined Paradigms

Modern Systems

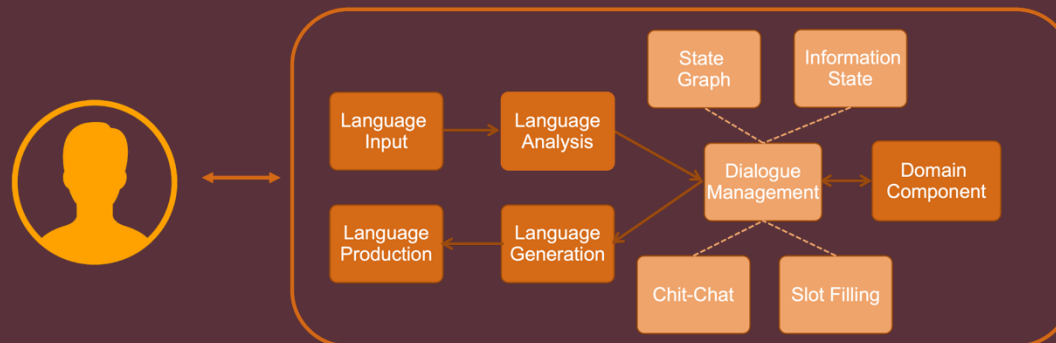
Dialogue Management

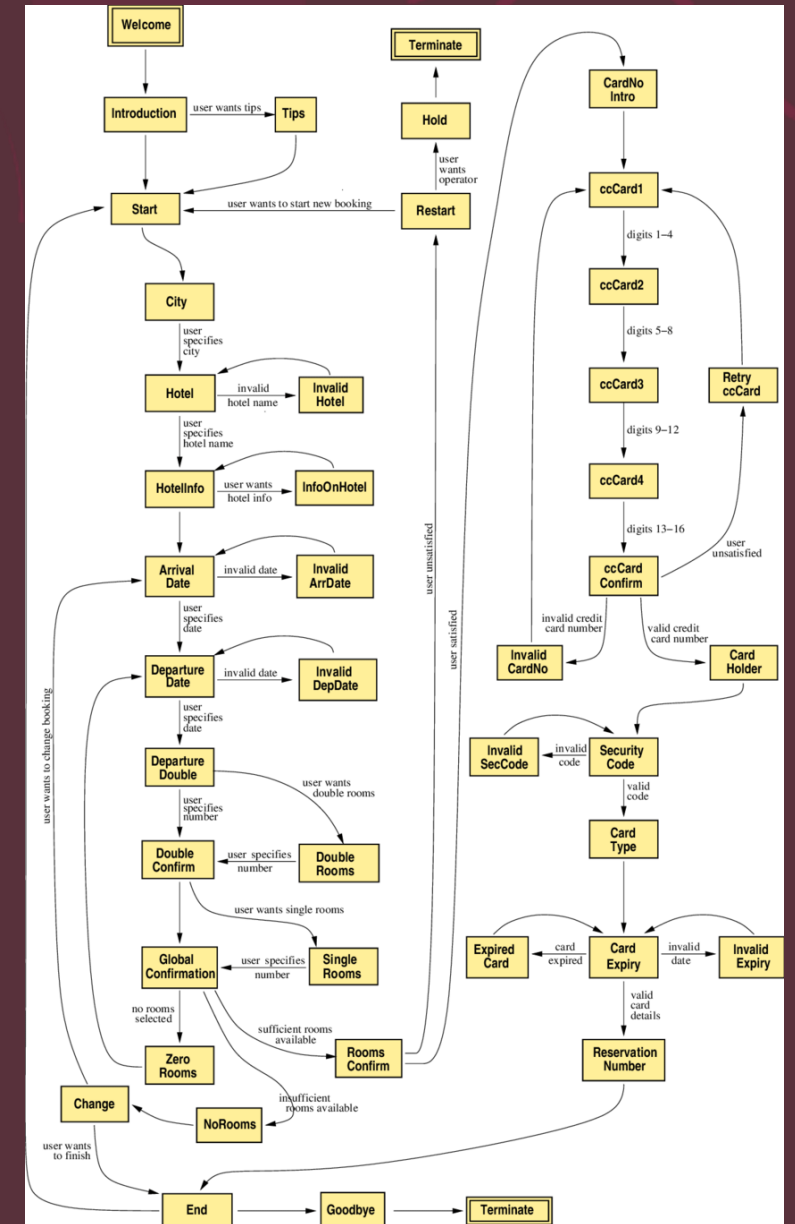
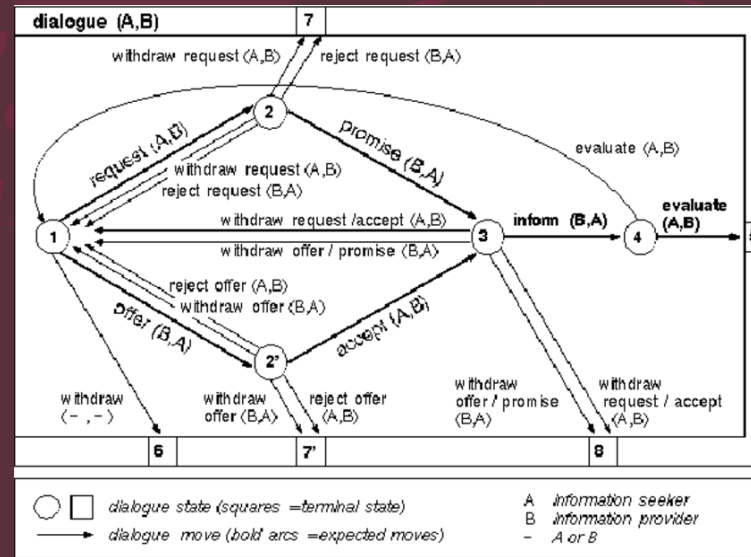


User	P	System
Hello	1.0	Hi, I'm pleased to meet you
What is your name?	1.0	My name is Stupido
I feel <emotion>	1.0	Oh, why do you think you feel <emotion>
Goodbye	0.5	Bye bye
Goodbye	0.5	Come back soon
....		
....		

Initiation Response (adjacency) Pairs with Probability and Slots

Chit-Chat Dialogue Management





FoodType
Burgers
Chinese
Korean
Sushi
....

Food	Day	Time	Location	Cost
Chinese			Dublin 7	cheap



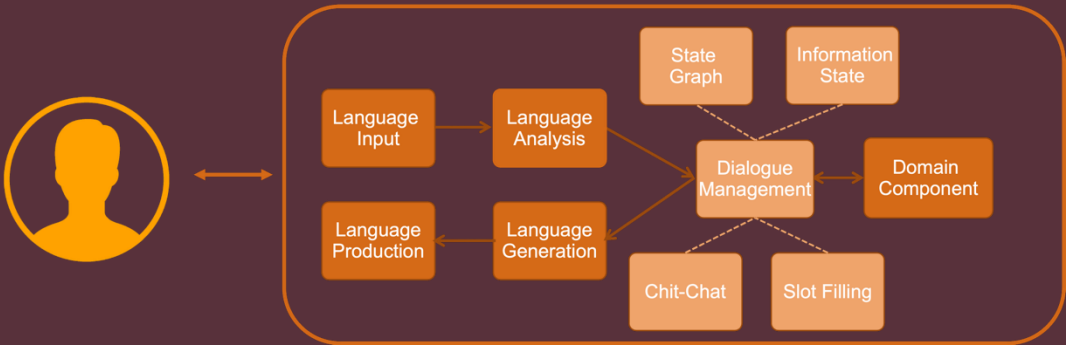
```

Algorithm Test Algorithm
1: procedure EUCLID(a, b)
2:   r ← a mod b
3:   while r ≠ 0 do
4:     a ← b
5:     b ← r
6:     r ← a mod b
7:   end while
8:   return b
% This line will not have a number!
% end procedure

```

▷ The g.c.d. of a and b
 ▷ We have the answer if r is 0
 ▷ The gcd is b

Slot Based Dialogue Management

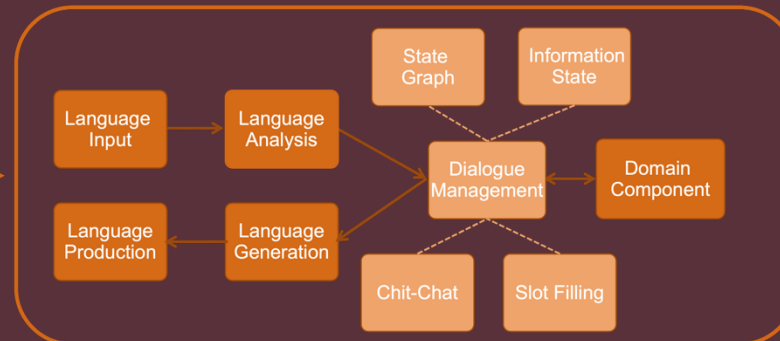


(RULE 3.1) **RULE: integrateUsrAsk**
CLASS: integrate

PRE: {
 \$/SHARED/LU/SPEAKER==usr
 fst(\$/PRIVATE/NIM, ask(Q))
 \$SCORE=Score
 Score > 0.7
 \$DOMAIN :: plan(Q, _Plan)

EFF: {
 1 pop(/PRIVATE/NIM)
 2 push(/PRIVATE/AGENDA, icm:acc*pos)
 3 add(/SHARED/LU/MOVES, ask(Q))
 4 if-do(Score ≤ 0.9,
 push(/PRIVATE/AGENDA, icm:und*pos:usr*issue(Q)))
 5 if-do(in(\$/SHARED/QUD, Q) and not fst(\$/SHARED/QUD, Q),
 push(/PRIVATE/AGENDA, icm:reraise:Q))
 6 push(/SHARED/QUD, Q)
 7 push(/PRIVATE/AGENDA, respond(Q))

Agent Based Dialogue Management



Dialogue Manager Paradigms ++

Slot Based
Dialogue Systems

State Graph
Dialogue Systems

ChatBot Based
(ChatGPT)

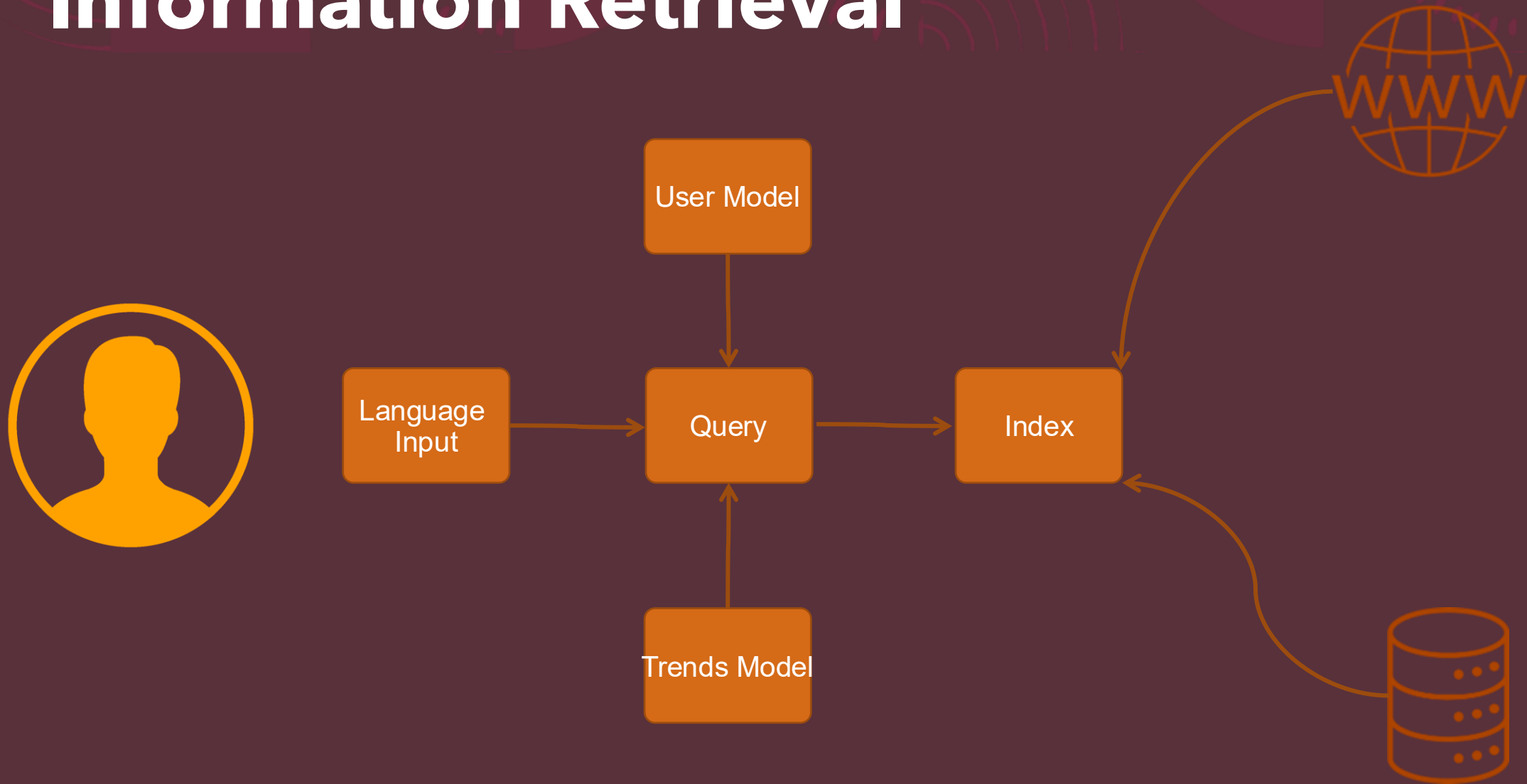
Agent Based
Dialogue Systems

Pull all 4 together
into Agentic LLM
solutions

Well Defined Paradigms

Modern Systems

Information Retrieval



The Basics

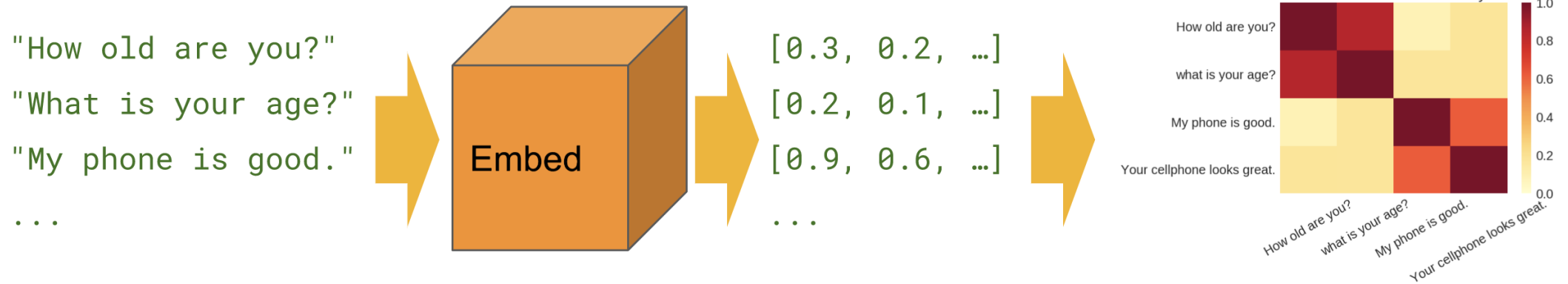
How do we
build
Conversational
Systems?

The ML
Playground

Putting Deep
Learning and
AI to Work

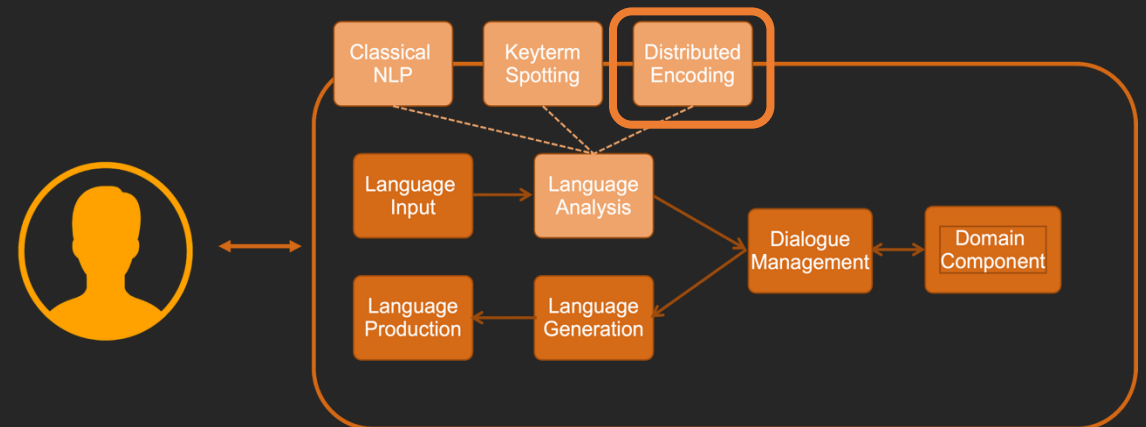
Personalisation

Keeping the
User Engaged

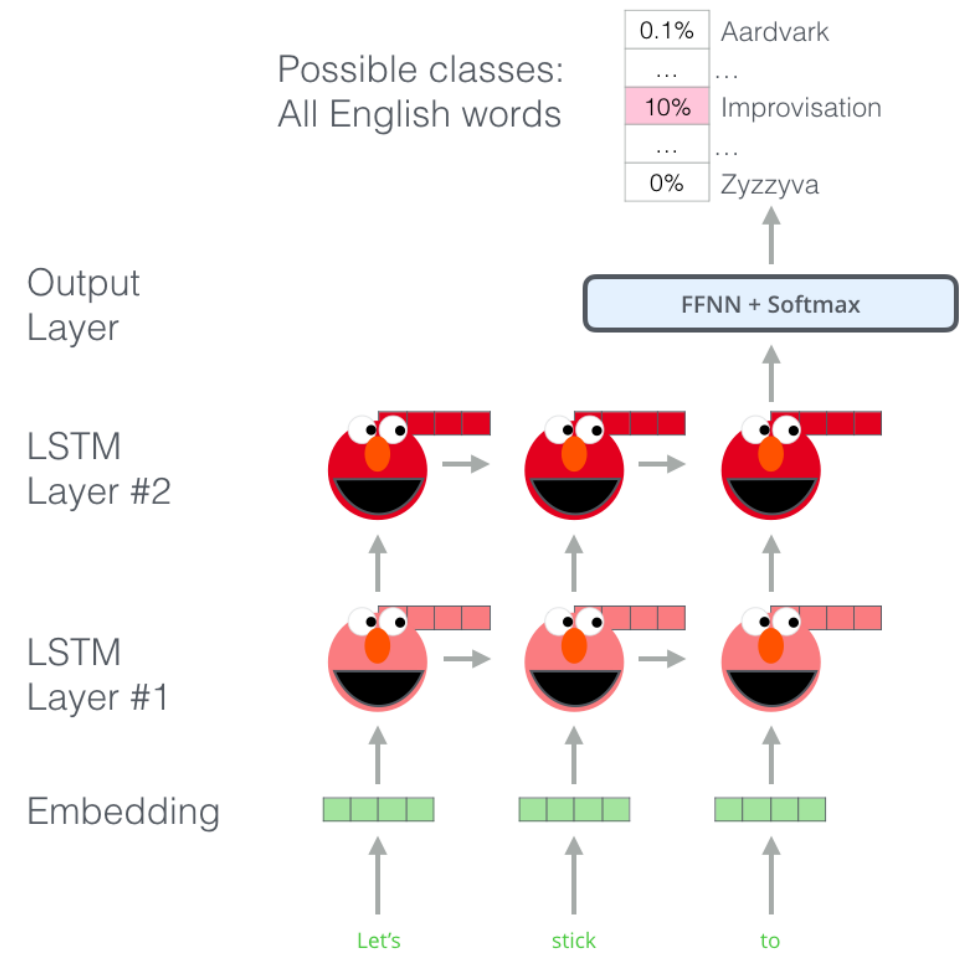
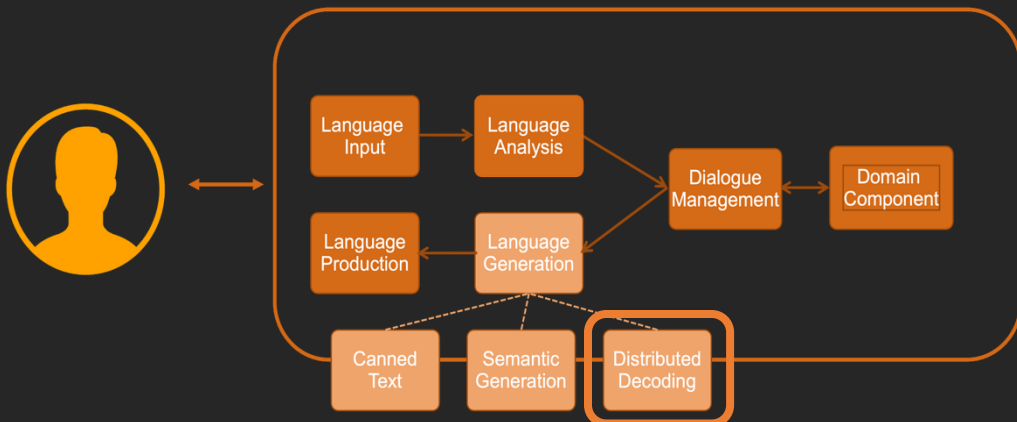


Tensorflow's Universal Sentence Encoder

Deep Learning at Work: Rethinking Understanding

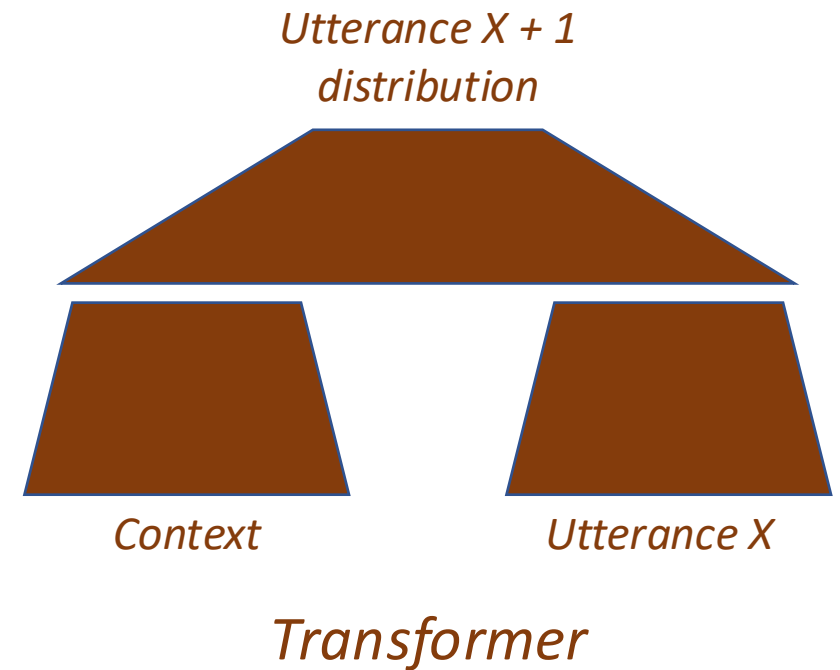
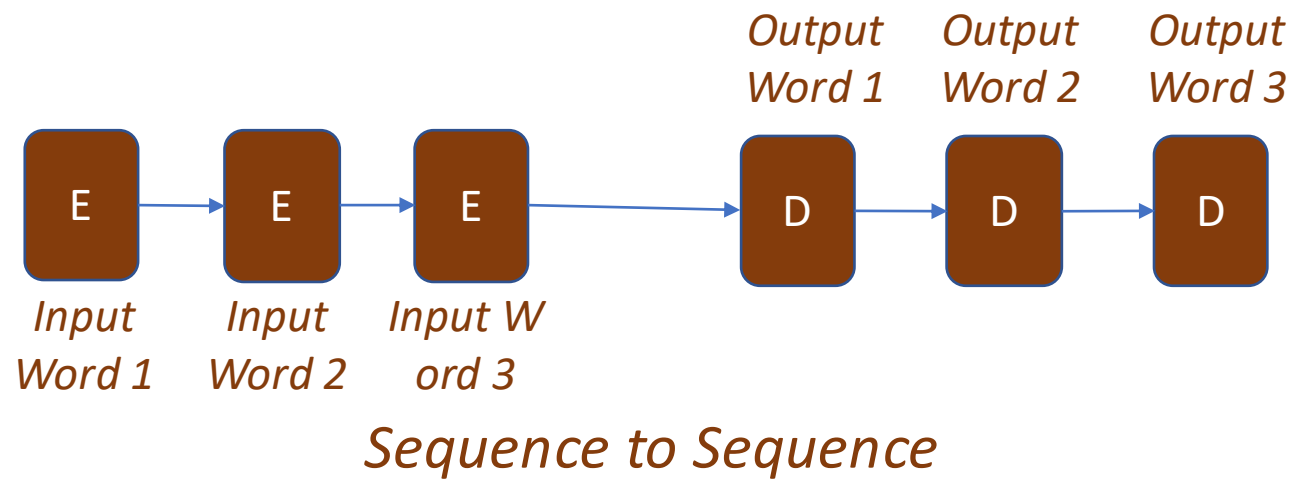


Deep Learning at work: Rethinking Language Production

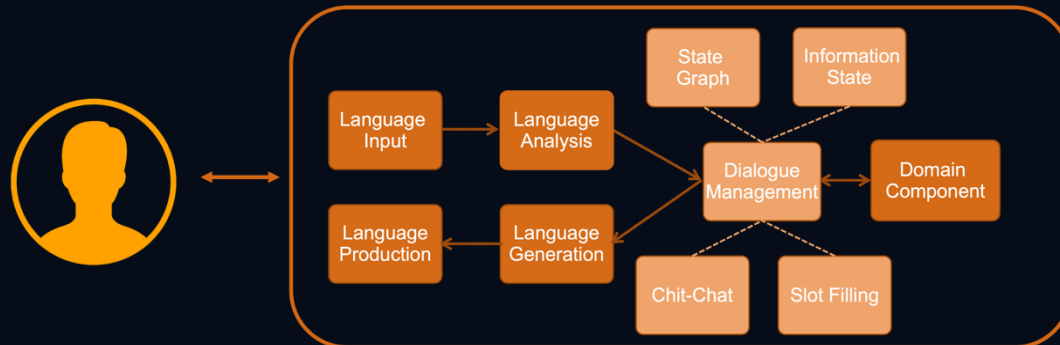


Sentence Decoders

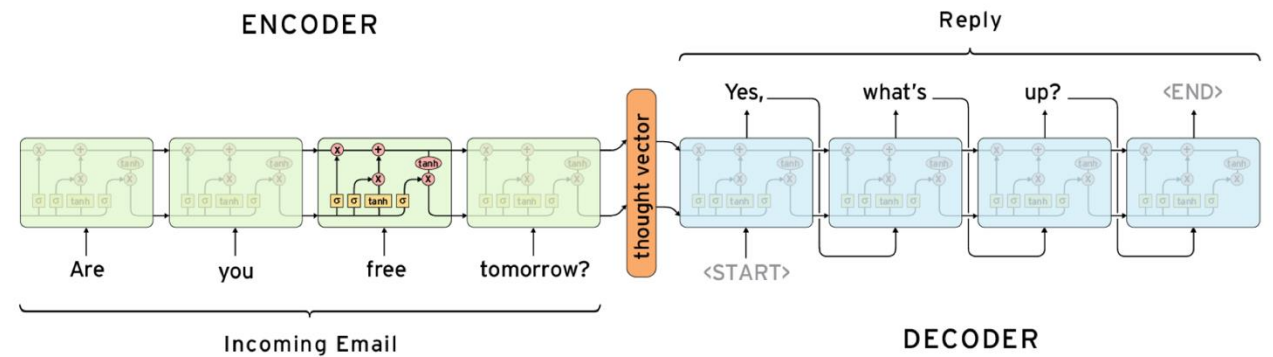
<https://jalammar.github.io/illustrated-bert/>



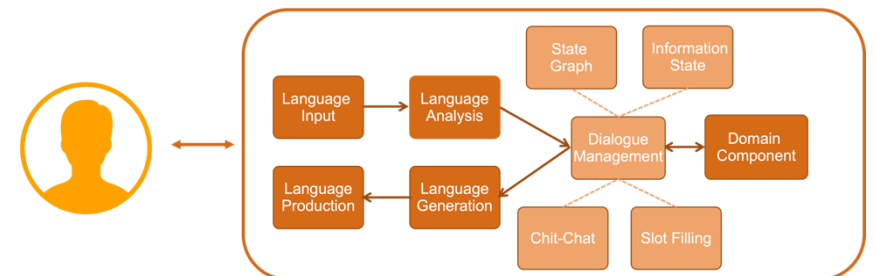
Deep Learning: Rethinking Dialogue Management



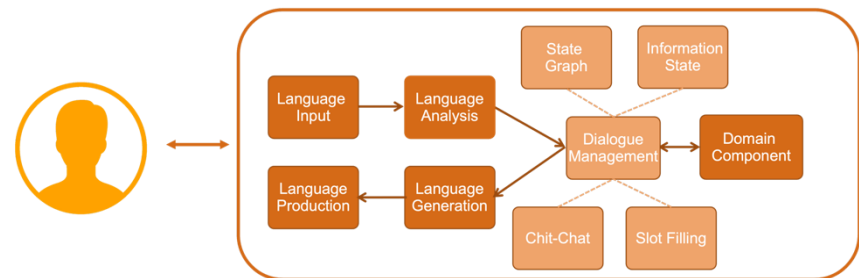
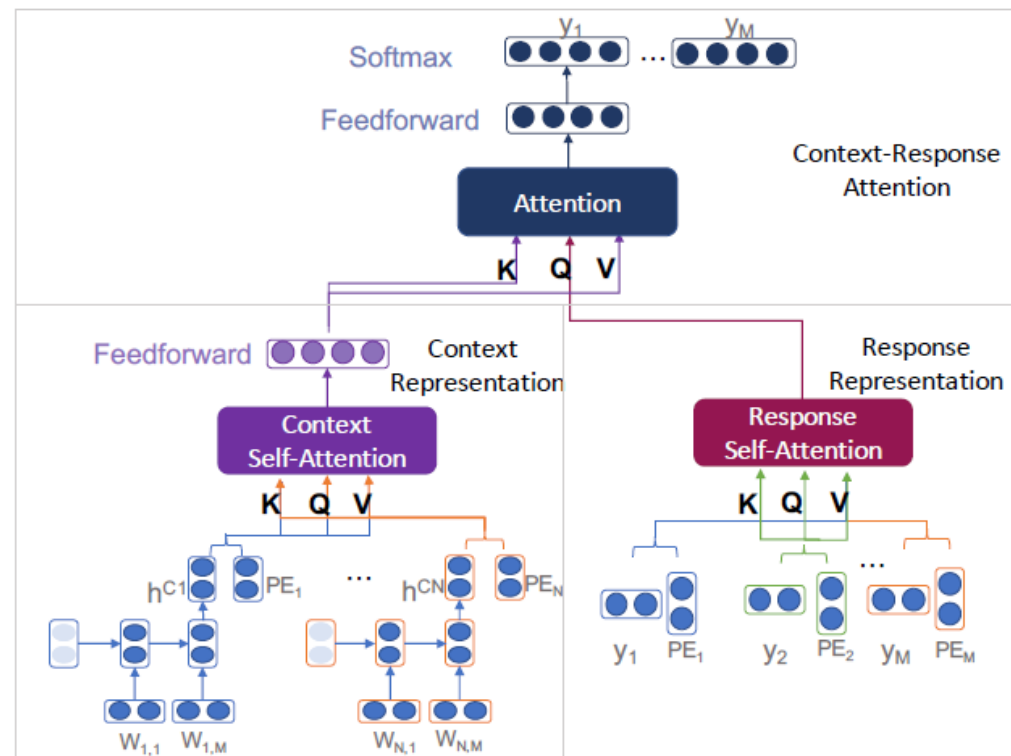
Pure DL Dialogue Management



Source <https://research.googleblog.com/2015/11/computer-respond-to-this-email.html>



Pure DL Dialogue Management



The Basics

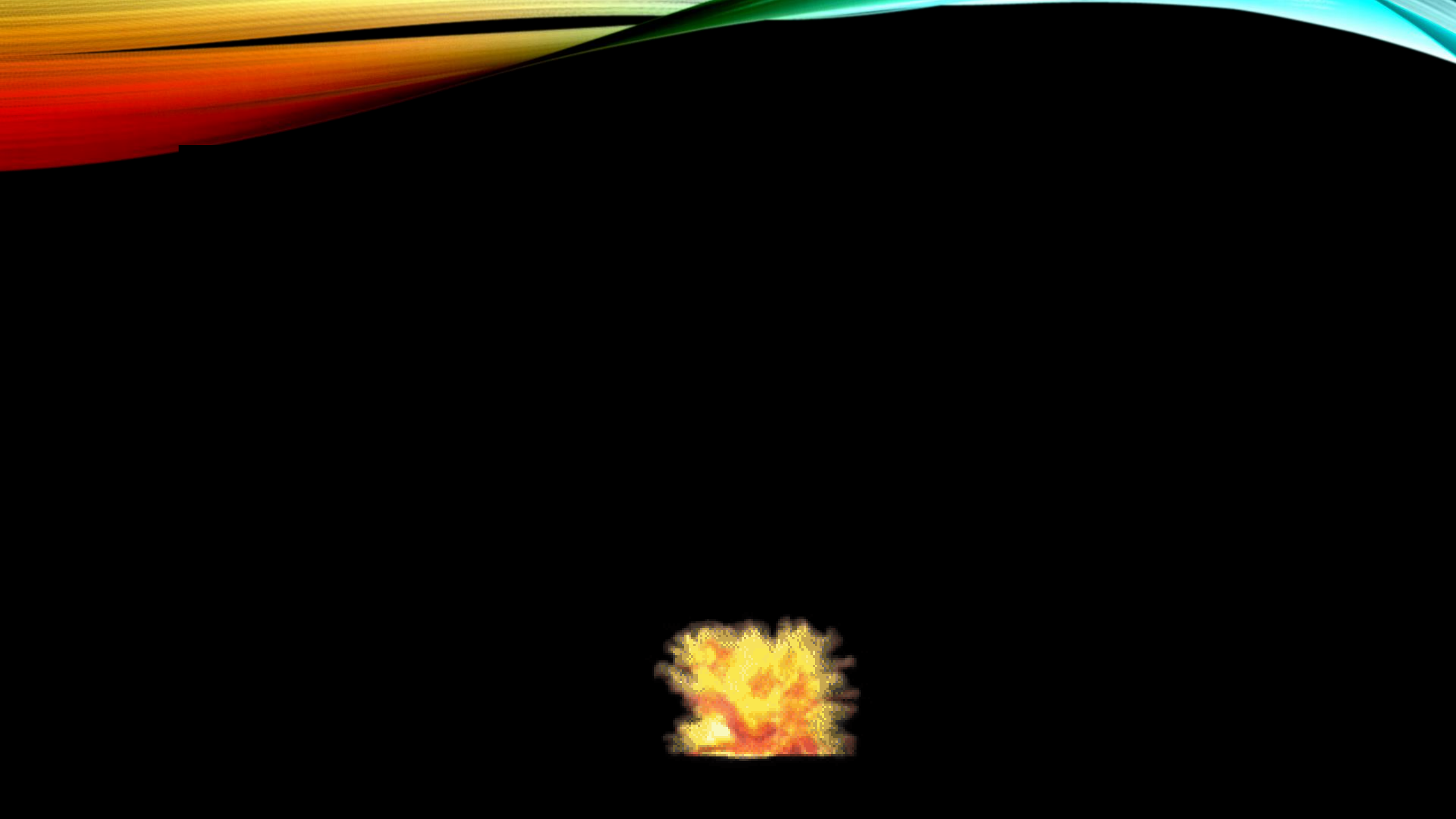
How do we
build
Conversational
Systems?

The ML
Playground

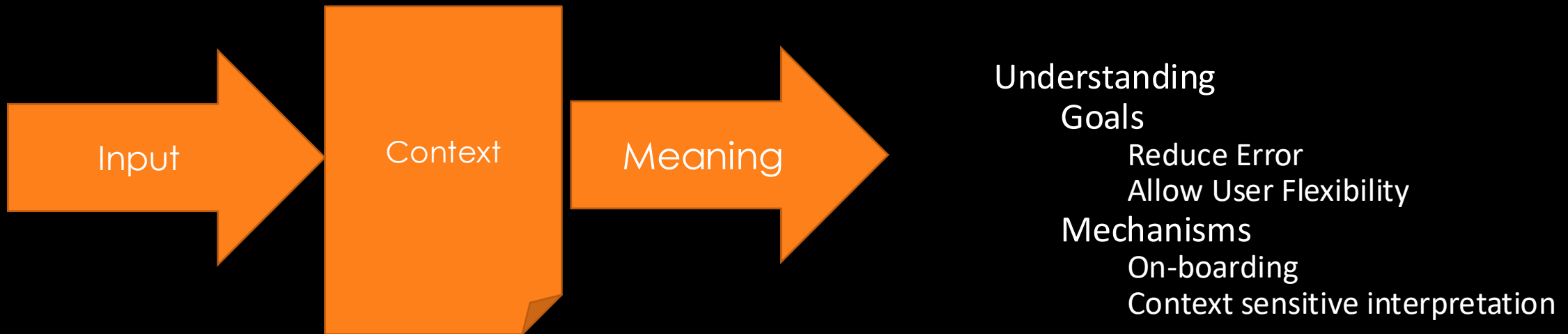
Putting Deep
Learning and
AI to Work

Personalisation

Keeping the
User Engaged



CONTEXTUALISATION



SHAPING TO THE USER

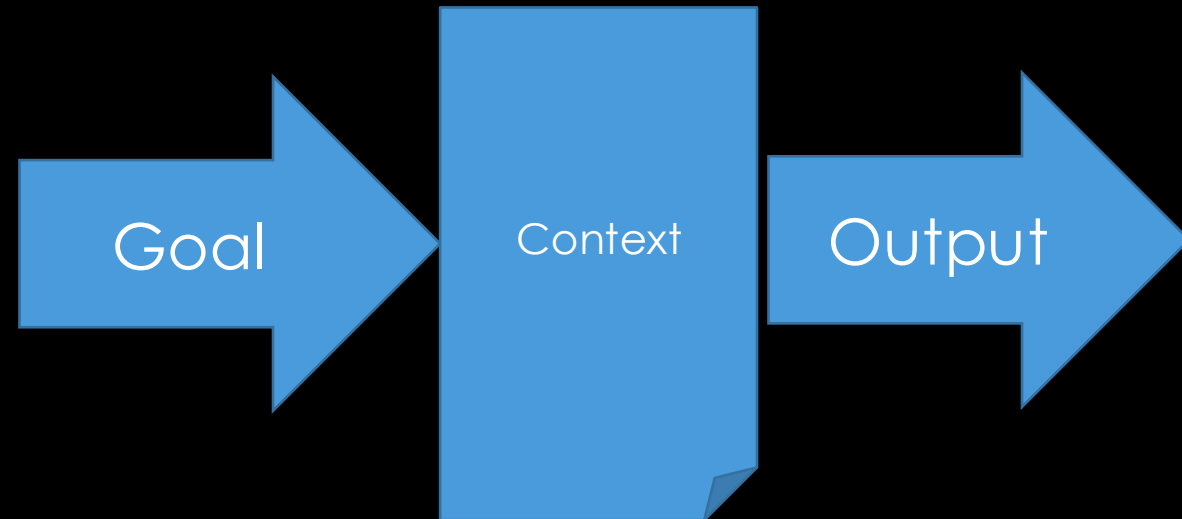
Production

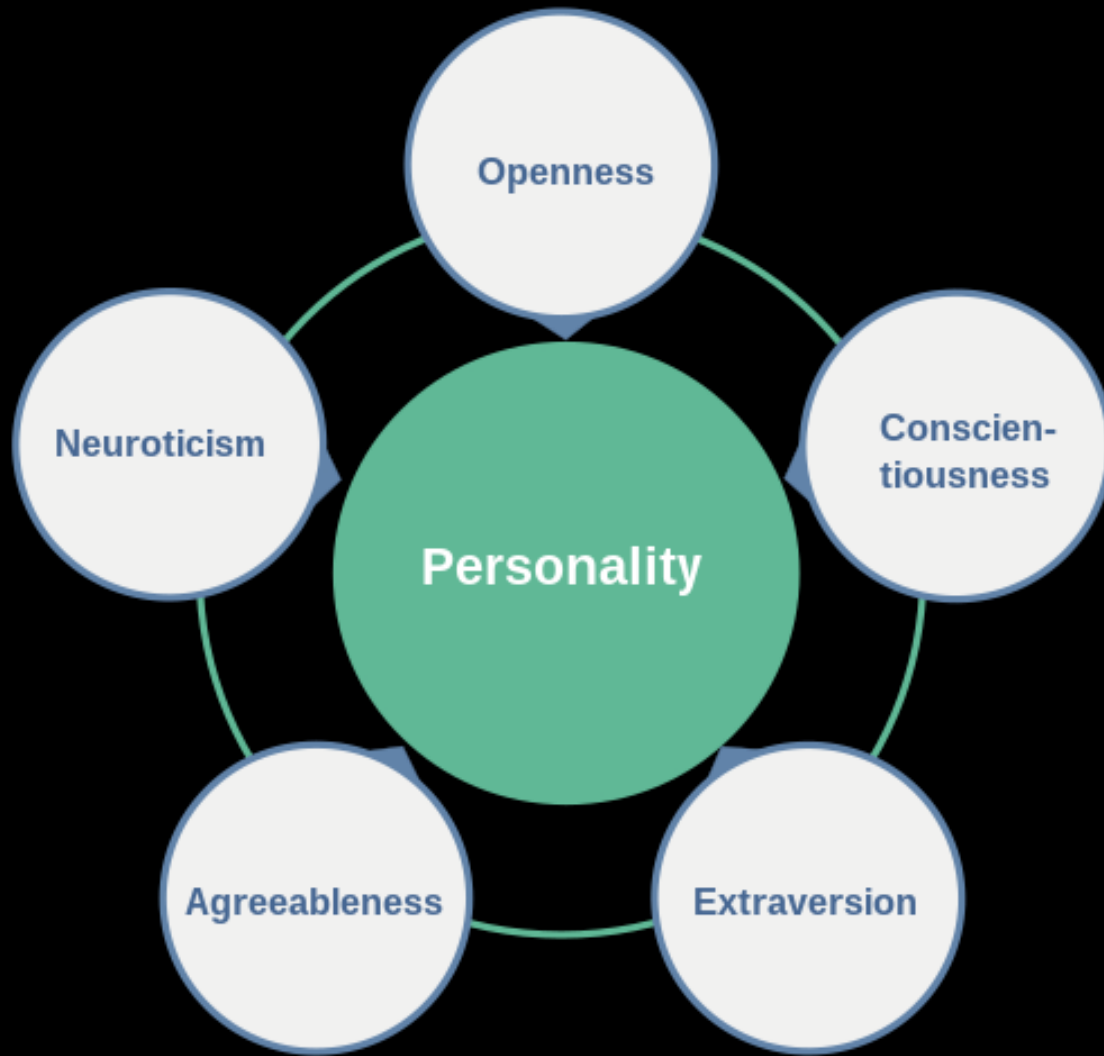
Goals

Improve Naturalness
Enhance satisfaction

Mechanisms

Dialogue planning
User state





PERSONALITY THE BIG FIVE

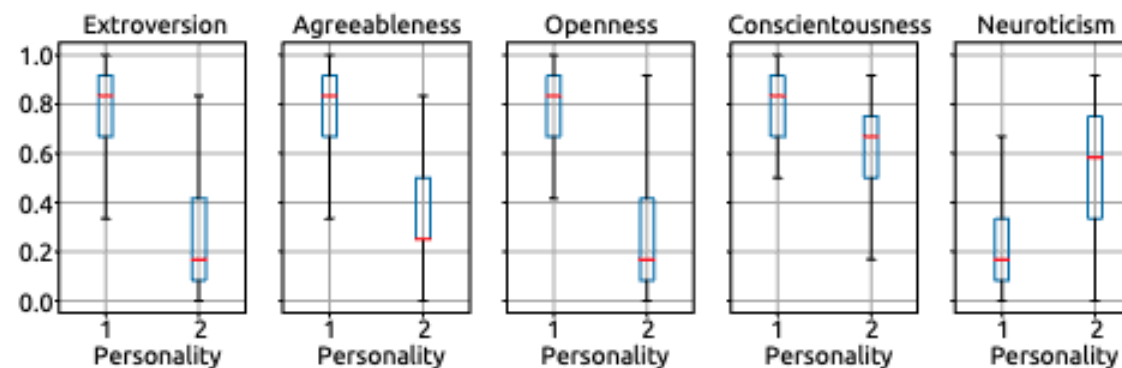


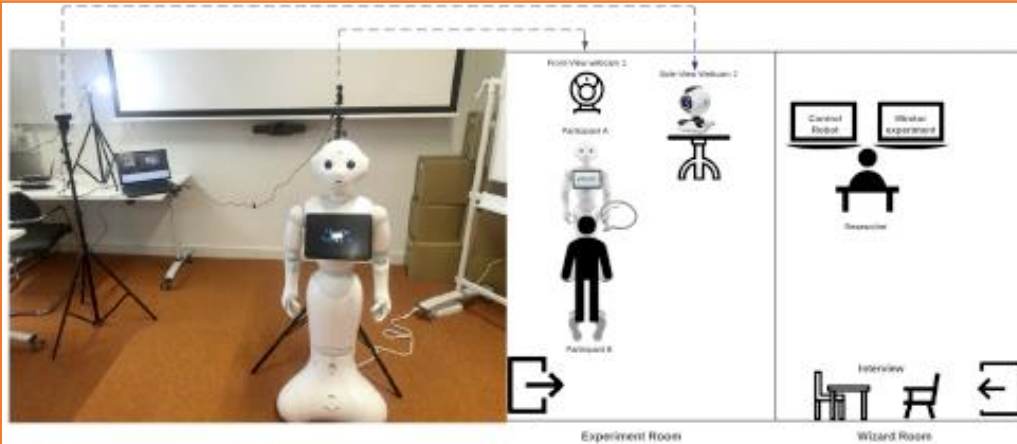
Virtual Tutor Personality in Computer Assisted Language Learning

Johanna Dobrinner



SFI centre for
research training in
digitally-enhanced reality





Confusion Detection in Interaction

by
Na Li



SFI Centre for Research Training
in Machine Learning





**Questions and
Challenges....**

GENERAL CHALLENGES



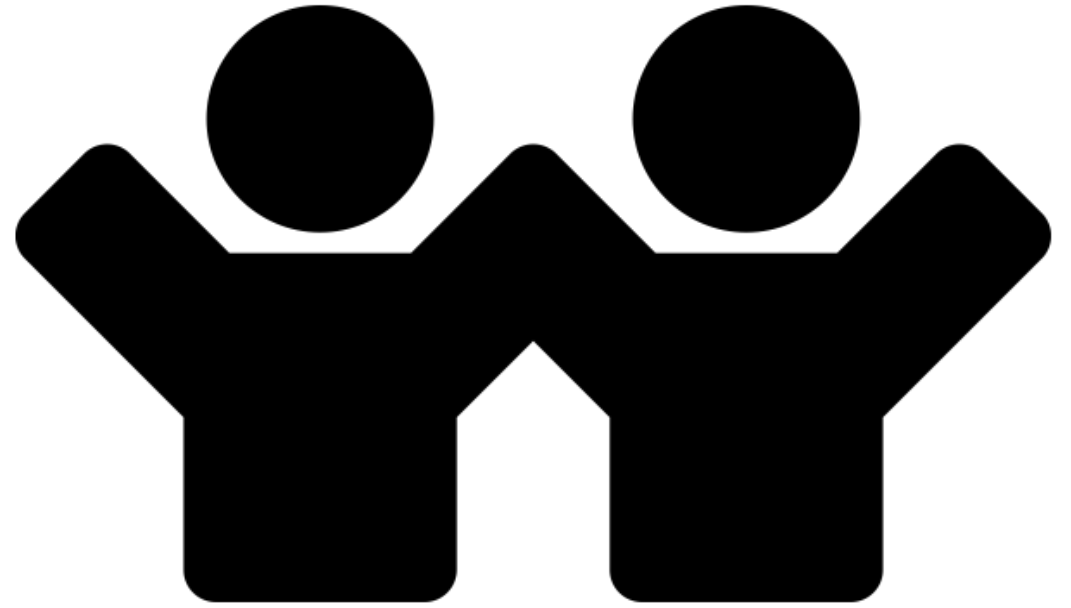
Usability and Acceptance



Sharing Data and Task
Structures



Standards in
Conversational Design



What role should conversational tech play?

Ethics



Privacy in Situated Conversational Systems

Systems immersed in a real or virtual environment which can perceive, reason on, and act on that environment





JOB

INTERVIEW
THIS WAY

